

EQUINESYNC PRODUCT IMPLEMENTATION ATLAS MASTER STANDARD AND TEMPLATE

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Related Implementation Authority: EquineSync MAIP

PART I

PRODUCT IMPLEMENTATION ATLAS MASTER STANDARD

1. Purpose

This Master Standard establishes the mandatory structure, quality requirements, lifecycle controls, review standards, traceability rules, evidence expectations, and reusable template for every EquineSync Product Implementation Atlas.

A Product Implementation Atlas, or PIA, is the authoritative domain-level bridge between EquineSync governance and executable software implementation.

The controlling implementation chain is:

**Constitutional Governance → Founder Decisions → MAIP → Product Implementation
Atlas → Engineering Work Packages → Code → Tests → Evidence → Release Decision →
Controlled Enrollment**

The purpose of this Standard is to ensure that EquineSync software is not built from incomplete assumptions, disconnected feature requests, informal conversations, or undocumented engineering decisions.

Every PIA must enable EquineSync to determine:

1. What must be built.
2. Why it must be built.
3. Who may use it.
4. Which data it owns or consumes.
5. How the capability behaves.
6. How failure is handled.
7. How the implementation is tested.
8. What evidence proves completion.
9. Whether the capability can be safely operated.
10. Whether it is ready for first-user enrollment.

2. Controlling Definition

The Founder has adopted the following definition:

A Product Implementation Atlas is the authoritative, version-controlled, domain-specific implementation instrument that translates EquineSync constitutional governance, MAIP requirements, founder decisions, product workflows, operational rules, data responsibilities, technical contracts, verification requirements, and release conditions into a complete and traceable basis for software construction and assurance.

This definition controls the drafting, review, approval, implementation use, validation, maintenance, and supersession of all EquineSync PIAs.

3. Normative Language

The following terms have mandatory meanings within this Standard:

Term	Meaning
MUST / SHALL	Mandatory requirement. Noncompliance blocks approval unless a formal deviation is authorized.
MUST NOT / SHALL NOT	Prohibited action or condition.
SHOULD	Strong recommendation. Any departure requires documented reasoning.

Term	Meaning
SHOULD NOT	Strong presumption against the action.
MAY	Permitted but not required.
CONTROLLED	Subject to documented authority, versioning, review, and change control.
AUTHORITATIVE	The recognized source of truth for the specified matter.
MATERIAL	Capable of changing functionality, user rights, safety, security, data integrity, legal posture, cost, implementation scope, or release readiness.
IMPLEMENTATION-READY	Complete enough for engineering to proceed without making unauthorized material decisions.
VERIFIED	Supported by objective test and evidence sufficient to demonstrate the requirement was satisfied.
FIRST-USER ENROLLMENT	The controlled admission and onboarding of the first non-founder user, facility, trainer, owner, rider, staff member, or other external participant into the operational EquineSync platform.

Use of weaker or ambiguous terms such as “generally,” “normally,” “as appropriate,” “user-friendly,” “secure,” “fast,” or “when needed” is prohibited unless the PIA defines an objective interpretation.

4. Authority and Order of Precedence

A PIA is subordinate to EquineSync constitutional governance and may not silently override, reinterpret, weaken, or conflict with a controlling source.

The order of authority is:

1. Locked EquineSync constitutional governance
2. Founder-adopted or founder-approved decisions
3. Applicable constitutional companion artifacts and authoritative registers
4. This PIA Master Standard
5. The controlling MAIP
6. Applicable approved domain and cross-domain PIAs
7. Engineering work packages
8. Source code, configuration, infrastructure, and migrations
9. Tests and validation scripts
10. Operational procedures and support materials

When a conflict is identified:

- The lower-authority source must not control.
- The conflict must be recorded as a finding.

- Implementation must pause for the affected requirement when proceeding could create material inconsistency.
- The matter must be resolved by the appropriate authority.
- The PIA must preserve the decision and resulting change history.

A PIA must not use implementation convenience as justification for disregarding governance.

5. Applicability

This Standard applies to:

- foundational platform PIAs
- product-domain PIAs
- cross-domain PIAs
- shared-service PIAs
- integration PIAs
- mobile implementation PIAs
- administrative and internal-operation PIAs
- enrollment and onboarding PIAs
- infrastructure or platform PIAs that materially affect user-facing behavior
- AI and automation PIAs
- reporting, analytics, and evidence PIAs
- implementation changes that materially alter an approved PIA

Examples include:

- Identity, Account, Actor, Enrollment, and Onboarding
- Facility and Organizational Structure
- Relationship, Authorization, and Permissions
- Horse Identity, Profile, and Lifecycle
- Health, Welfare, and Medical Records
- Care Operations
- Feed and Nutrition
- Scheduling and Calendar
- Lessons, Training, Riders, and Guardians
- Shows, Events, Ride Times, and Trainer Itineraries
- Billing, Payments, and Financial Operations
- Owner Portal
- Communications and Notices
- Files, Media, and Digital Assets
- Search and Discovery
- Reporting and Analytics
- AI and Automation
- Mobile and Offline Experience
- Integrations and Developer Platform
- Platform Administration and Support

A PIA may cover more than one closely related domain only when:

- the domains cannot be implemented coherently in isolation;
 - ownership boundaries remain explicit;
 - the combined scope remains reviewable; and
 - the combined document does not conceal unresolved cross-domain conflicts.
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6. PIA Quality Standard

Every PIA must satisfy the EquineSync **BRAVO** quality standard.

6.1 Buildable

Engineering can implement the capability without inventing material requirements, permissions, workflows, data rules, or release conditions.

6.2 Reviewable

A segregated reviewer can determine whether the proposed implementation is complete, internally consistent, and compliant with controlling sources.

6.3 Auditable

Every material requirement, decision, exception, implementation result, and release disposition can be traced to an authoritative source and preserved evidence.

6.4 Verifiable

Quality assurance and machine validation can derive objective tests from the PIA and determine whether the implementation satisfies it.

6.5 Operable

EquineSync can deploy, monitor, support, recover, correct, maintain, and safely retire the capability.

A PIA that fails any BRAVO element is not implementation-ready.

7. Core Drafting Principles

Every PIA must follow these principles.

7.1 No Silent Invention

Material product, security, permission, data, legal, operational, or release decisions must not be invented by engineering, drafting agents, reviewers, or automated systems.

Unresolved material questions must be recorded and escalated.

7.2 One Authoritative Owner Per Record or Decision

Every material record type, state, and decision must have one authoritative owner.

Other domains may reference, cache, display, transform, or derive information only under defined rules.

7.3 Whole-Workflow Specification

A PIA must define the complete workflow, not only the primary screen or happy path.

It must include:

- initiation
- validation
- authorization
- intermediate states
- completion
- failure
- retry
- correction
- cancellation
- reversal
- archival
- audit
- notification
- support and recovery

7.4 Least Privilege

Access must be limited to the minimum authority necessary for the actor, relationship, facility, horse, record, and purpose.

7.5 Evidence by Design

Evidence requirements must be defined before implementation begins.

7.6 Operational Realism

PIAs must account for actual equine environments, including:

- mobile use
- gloves and one-handed use
- poor connectivity
- outdoor glare
- urgent care situations
- shared responsibilities
- changing staff
- temporary delegation
- last-minute show changes
- multiple horses and riders
- multiple facilities
- incomplete information
- time-sensitive communications

7.7 Explicit Release Boundaries

The PIA must distinguish what is required for:

- internal build
- founder pilot
- first-user enrollment
- paid enrollment
- production launch
- later enhancement

7.8 Human-Readable and Machine-Verifiable

The PIA must remain understandable to the Founder, domain reviewers, operational users, engineers, and assurance agents while also supporting structured validation.

8. PIA Classification Types

Each PIA must be classified as one or more of the following:

Type	Purpose
Foundational PIA	Defines capabilities required by most other domains, such as identity, tenancy, relationships, permissions, audit, or application shell.
Domain PIA	Defines one primary EquineSync business or product domain.
Cross-Domain PIA	Defines a workflow or service spanning multiple authoritative domains.

Type	Purpose
Platform PIA	Defines infrastructure, deployment, observability, feature flags, resilience, or shared technical services.
Integration PIA	Defines external system, API, import, export, webhook, or adapter behavior.
Experience PIA	Defines a coherent user experience across multiple capabilities, such as mobile, owner portal, onboarding, or accessibility.
Migration PIA	Defines controlled data import, conversion, reconciliation, rollback, and verification.
Assurance PIA	Defines evidence, validation, testing, review, or release-control capabilities.

The classification must not be used to avoid required sections. A section may be marked not applicable only with a documented basis.

9. Required Roles and Segregation

Each PIA must identify the named person, function, or controlled agent responsible for each role.

Role	Minimum Responsibility
Founder / Approval Authority	Approves material scope, unresolved decisions, retained risks, and implementation authorization.
PIA Owner	Accountable for completeness, maintenance, and cross-domain coordination.
Drafting Function	Produces the PIA from controlling sources without unauthorized invention.
Domain Reviewer	Validates operational accuracy and domain completeness.
Architecture Reviewer	Validates technical coherence and boundary alignment.
Security and Privacy Reviewer	Validates access, data protection, consent, and abuse controls.
Segregated Reviewer	Independently reviews work not authored by that reviewer.
Adversarial Reviewer	Challenges assumptions, failure handling, misuse cases, and hidden gaps.
Machine Validation Function	Validates schemas, identifiers, references, required sections, and structured rules.
Golden-Path Reproduction Function	Reproduces representative end-to-end workflows.

Role	Minimum Responsibility
Evidence Custodian	Preserves, indexes, verifies, and packages evidence.
Engineering Owner	Converts the approved PIA into controlled engineering work.
QA Owner	Derives and executes verification from acceptance criteria.
Operational Owner	Confirms support, monitoring, recovery, and maintenance readiness.

One person may perform multiple roles where staffing is limited, but the PIA must disclose the overlap.

The following functions must remain substantively segregated whenever feasible:

- drafting and independent review
- implementation and final verification
- evidence production and evidence acceptance
- risk creation and risk acceptance

Where full personnel segregation is impossible, EquineSync must use procedural segregation, separate review passes, distinct prompts, independent evidence review, and Founder acknowledgment.

10. PIA Lifecycle

Every PIA must use a controlled lifecycle.

10.1 Lifecycle States

State	Meaning
NOT_STARTED	No controlled drafting has begun.
SOURCE_ASSEMBLY	Controlling sources are being gathered and verified.
DRAFTING	The PIA is under active drafting.
DRAFT_COMPLETE	The initial complete draft exists.
DOMAIN_REVIEW	Operational and domain review is underway.
SEGREGATED_REVIEW	Independent review is underway.
ADVERSARIAL_CHALLENGE	Failure, misuse, ambiguity, and completeness are being challenged.
MACHINE_VALIDATION	Structured validation and reference checks are underway.
FOUNDER_REVIEW	The candidate is presented for Founder decision.

State	Meaning
FOUNDER_APPROVED	The PIA content is approved but implementation may not yet be authorized.
IMPLEMENTATION_AUTHORIZED	Engineering may build against the approved baseline.
IN_IMPLEMENTATION	Controlled implementation is underway.
IMPLEMENTED_PENDING_VERIFICATION	Implementation is complete but not fully verified.
VERIFICATION_IN_PROGRESS	Testing and evidence collection are underway.
VERIFIED_WITH_FINDINGS	Verification completed with unresolved findings.
VERIFIED_RELEASE_CANDIDATE	Required verification passed for the stated release scope.
READY_FOR_FIRST_USER_ENROLLMENT	All enrollment gates for the applicable scope are satisfied.
ACTIVE	The PIA controls an operational production capability.
REVISION_REQUIRED	Material correction or expansion is required.
BLOCKED	Progress is blocked by a dependency, decision, defect, or missing source.
SUPERSEDED	A later approved version controls.
WITHDRAWN	The PIA is no longer authorized for use.
ARCHIVED	Retained for historical or evidentiary purposes.

10.2 Prohibited Status Inflation

A PIA must not be labeled:

- approved when only drafted;
- implemented when only partially coded;
- verified when only reviewed;
- release-ready when evidence is missing;
- enrollment-ready when operational or onboarding controls remain incomplete.

The status must represent the least-complete material component.

11. Versioning and Change Control

Each PIA must use semantic document versioning:

- **Major version:** Material scope, architecture, data ownership, permissions, lifecycle, or release change
- **Minor version:** New capability, expanded workflow, or nonbreaking substantive change

- **Patch version:** Clarification, correction, formatting, reference repair, or non-substantive refinement

Example:

ES-PIA-HORSE-LIFECYCLE-V1.2.1

Every version must include:

- version number
- status
- date
- author or drafting function
- reviewers
- source baseline
- change summary
- affected requirements
- affected tests
- affected evidence
- migration implications
- prior version
- supersession status

A material change to an approved PIA requires impact assessment before implementation.

12. Identifier Standard

Every material PIA element must have a stable unique identifier.

12.1 PIA Identifier

Format:

ES-PIA-[DOMAIN]-V[MAJOR].[MINOR].[PATCH]

Example:

ES-PIA-IDENTITY-ONBOARDING-V1.0.0

12.2 Element Identifiers

Element	Format	Example
Requirement	[DOMAIN]-REQ-###	HORSE-REQ-014
Founder Decision	[DOMAIN]-FD-###	HORSE-FD-003

Element	Format	Example
Assumption	[DOMAIN]-ASM-###	HORSE-ASM-002
Open Question	[DOMAIN]-OQ-###	HORSE-OQ-006
Workflow	[DOMAIN]-WF-###	HORSE-WF-004
Business Rule	[DOMAIN]-BR-###	HORSE-BR-019
Entity	[DOMAIN]-ENT-###	HORSE-ENT-001
State Model	[DOMAIN]-SM-###	HORSE-SM-002
Permission Rule	[DOMAIN]-PERM-###	HORSE-PERM-011
Interface Requirement	[DOMAIN]-UI-###	HORSE-UI-008
API Contract	[DOMAIN]-API-###	HORSE-API-005
Event	[DOMAIN]-EVT-###	HORSE-EVT-003
Integration	[DOMAIN]-INT-###	HORSE-INT-002
Acceptance Criterion	[DOMAIN]-AC-###	HORSE-AC-027
Test Case	[DOMAIN]-TEST-###	HORSE-TEST-041
Evidence Item	[DOMAIN]-EVID-###	HORSE-EVID-018
Dependency	[DOMAIN]-DEP-###	HORSE-DEP-007
Risk	[DOMAIN]-RISK-###	HORSE-RISK-004
Finding	[DOMAIN]-FIND-[P0/P1/P2]-###	HORSE-FIND-P1-002
Deviation	[DOMAIN]-DEV-###	HORSE-DEV-001
Migration Rule	[DOMAIN]-MIG-###	HORSE-MIG-009
Metric	[DOMAIN]-METRIC-###	HORSE-METRIC-005

Identifiers must not be reused after deletion or supersession.

13. Mandatory PIA Sections

Every PIA must contain the following sections in the stated order unless this Standard expressly authorizes consolidation.

1. Document Control and Status
2. Executive Summary
3. Purpose and Intended Outcome
4. Authoritative Sources

5. Domain Scope and Boundaries
6. Definitions and Controlled Vocabulary
7. Actors, Roles, Relationships, and Authorities
8. Capability Map
9. User and Operational Workflows
10. Business Rules
11. Data Entities and Relationships
12. Record Ownership and Stewardship
13. State and Transition Models
14. Authorization and Permission Matrix
15. User Interface and Experience Requirements
16. API, Event, Job, and Integration Contracts
17. Notifications and Communications
18. Files, Media, and Document Handling
19. Search, Reporting, and Analytics Requirements
20. Offline and Synchronization Behavior
21. Security, Privacy, Consent, and Safeguarding
22. Failure Modes, Recovery, and Correction
23. Observability, Administration, and Support Operations
24. Nonfunctional Requirements
25. Migration, Configuration, Feature Flags, and Seed Data
26. Acceptance Criteria
27. Test and Validation Matrix
28. Golden-Path Reproduction Scenarios
29. Adversarial, Negative, and Abuse Scenarios
30. Evidence Requirements and Evidence Manifest
31. Dependencies and Implementation Sequence
32. Release Scope and Readiness Gates
33. Open Decisions, Findings, Assumptions, and Retained Risks
34. Change-Control History
35. Requirement Traceability Matrix
36. Five Mandatory Readiness Questions
37. Review, Approval, and Implementation Authorization

No required section may be omitted. Where a section is not applicable, the PIA must state:

- `NOT_APPLICABLE`
- the reason
- the reviewing authority that accepted the determination

14. Requirement Drafting Standard

Every requirement must be:

- singular
- clear

- testable
- attributable
- bounded
- implementation-relevant
- free from hidden assumptions

Each requirement must include:

Field	Required Content
Requirement ID	Stable unique identifier
Requirement Statement	One normative obligation
Source	Controlling authority
Rationale	Why the requirement exists
Actor or System	Who or what performs it
Preconditions	Conditions required before execution
Expected Behavior	Required outcome
Prohibited Behavior	What must not occur
Failure Behavior	What happens when execution fails
Release Classification	When it is required
Acceptance Criteria	Objective proof standard
Test References	Linked test cases
Evidence References	Linked evidence items
Status	Draft, approved, implemented, verified, deferred, or superseded

14.1 Example of a Strong Requirement

SHOW-REQ-021

The system SHALL prevent publication of a show schedule containing overlapping ride assignments for the same horse or rider unless an authorized show administrator records and accepts an explicit conflict override.

This requirement is objectively testable and identifies the prohibited state.

14.2 Weak Requirement

The system should make scheduling easy.

This is prohibited because it does not define measurable behavior.

15. Source Traceability

Every PIA must include a source register.

Sources must be classified as:

- controlling constitutional authority
- founder decision
- MAIP requirement
- approved predecessor
- authoritative technical standard
- domain evidence
- implementation assumption
- non-controlling reference

Every material PIA requirement must trace to at least one source.

Every source-derived obligation must trace forward to:

- PIA requirement
- implementation work package
- code or configuration
- test
- evidence
- release gate

No orphaned requirement, test, or evidence item is permitted.

16. Domain Boundaries and Ownership

Each PIA must identify:

- what the domain owns
- what the domain references
- what the domain may derive
- what the domain may display
- what the domain may modify
- what the domain must never modify
- incoming dependencies
- outgoing dependencies
- cross-domain conflicts
- authority for conflict resolution

Each material record must have:

- authoritative system
- authoritative domain
- record steward
- permitted creators
- permitted editors
- retention authority
- deletion authority
- correction authority
- audit requirements

A displayed copy must not be mistaken for the authoritative record.

17. Workflow Standard

Each material workflow must define:

1. Workflow identifier and name
2. Purpose
3. Actors
4. Preconditions
5. Trigger
6. Required inputs
7. Optional inputs
8. Authorization checks
9. Ordered steps
10. State changes
11. Records created or modified
12. Notifications
13. Audit events
14. Success outcome
15. Partial-success behavior
16. Failure behavior
17. Retry behavior
18. Cancellation behavior
19. Correction behavior
20. Reversal behavior
21. Offline behavior
22. Accessibility requirements
23. Acceptance criteria
24. Test cases
25. Evidence

Flow diagrams may supplement but must not replace normative text.

18. State and Transition Standard

Every material lifecycle must have a state model.

Each state model must define:

- permitted states
- initial state
- terminal states
- transition triggers
- authorized actors
- required fields
- validation
- audit events
- notifications
- reversible transitions
- prohibited transitions
- timeout behavior
- suspension behavior
- expiration behavior
- correction process

State transitions must not depend solely on user-interface controls. Server-side enforcement is required for material rules.

19. Authorization and Permission Standard

Every PIA must include a permission matrix that accounts for:

- actor identity
- account status
- role
- relationship
- facility
- organization
- horse
- record type
- action
- purpose
- delegation
- guardian authority
- consent
- time limitation
- emergency authority
- record sensitivity
- platform administration

For each action, the PIA must define:

- who may perform it
- what relationship authorizes it
- what conditions limit it
- when authority begins
- when authority ends
- whether it may be delegated
- whether approval is required
- whether the action is logged
- whether the affected person is notified
- whether the action is reversible

Permission enforcement must be consistent across:

- web
- mobile
- API
- search
- export
- reporting
- notifications
- automation
- AI features
- integrations
- internal administration

A hidden button is not an authorization control.

20. Data Model Standard

Each material entity must include:

- entity identifier
- name
- purpose
- authoritative owner
- fields
- field types
- required status
- validation
- relationships
- cardinality
- uniqueness rules
- indexing requirements
- sensitivity classification

- retention
- archival
- deletion
- correction
- audit
- searchability
- exportability
- offline availability
- provenance
- derivation rules

The PIA must distinguish:

- authoritative values
- user observations
- imported values
- calculated values
- inferred values
- cached values
- AI-generated suggestions
- historical snapshots
- immutable evidence
- superseded values

High-risk records must preserve provenance and correction history.

21. User Interface and Experience Standard

UI requirements must define:

- supported platforms
- supported roles
- primary tasks
- navigation entry points
- required fields
- optional fields
- validation
- empty states
- loading states
- offline states
- error states
- permission-denied states
- confirmation behavior
- destructive-action safeguards
- accessibility
- responsive behavior

- mobile behavior
- autosave behavior
- progress preservation
- support links
- user-visible audit or status information

Every important action must clearly communicate:

- what happened
 - whether it succeeded
 - what remains pending
 - whether synchronization is required
 - whether another person must act
 - how to correct a mistake
-

22. API, Event, Job, and Integration Standard

The PIA must identify each required:

- command
- query
- API endpoint
- domain event
- webhook
- background job
- scheduled job
- import
- export
- third-party adapter
- retry queue
- reconciliation process

Each contract must define:

- contract identifier
- purpose
- caller
- authorization
- request data
- response data
- validation
- error codes
- idempotency
- rate limits
- retry behavior
- timeout behavior

- versioning
- audit behavior
- privacy constraints
- observability
- test requirements

External systems must not become an undeclared source of truth.

23. Notifications and Communications Standard

Each notification must define:

- event trigger
- sender identity
- recipient
- channel
- required content
- urgency
- consent basis
- delivery timing
- retry behavior
- failure handling
- duplication prevention
- read or acknowledgment behavior
- localization or time-zone treatment
- audit evidence
- preference controls
- mandatory notice exceptions

The PIA must distinguish:

- operational notification
 - legal or governance notice
 - marketing communication
 - emergency alert
 - task reminder
 - system status message
 - human-authored message
-

24. Offline and Synchronization Standard

Each applicable PIA must specify:

- functions available offline
- records available offline

- storage duration
- device protection
- queued actions
- actions prohibited offline
- synchronization trigger
- conflict detection
- conflict resolution
- stale-data warnings
- retry rules
- user-visible sync status
- duplicate prevention
- data-loss safeguards
- remote revocation behavior
- prolonged-offline behavior
- support diagnostics

Offline capability must be evaluated per workflow.

The phrase “offline supported” is insufficient without defined behavior.

25. Security, Privacy, Consent, and Safeguarding Standard

Security and privacy controls must be embedded in each workflow.

The PIA must address, where applicable:

- authentication
- session control
- authorization
- least privilege
- encryption
- data classification
- credential handling
- consent
- legal authority
- guardian authority
- minor safeguarding
- audit logging
- suspicious activity
- abuse prevention
- data minimization
- purpose limitation
- retention
- deletion restrictions
- export restrictions
- support access

- emergency access
- secrets management
- vendor access
- incident response
- breach evidence
- AI data use

The PIA must identify prohibited uses of sensitive data.

26. AI and Automation Requirements

Any PIA using AI, automated recommendations, predictions, classifications, extraction, generation, or autonomous action must define:

- intended use
- prohibited use
- data inputs
- data provenance
- model or provider boundary
- confidence handling
- human review
- explainability
- correction
- audit
- feedback
- hallucination risk
- bias risk
- safety limits
- fallback behavior
- model failure behavior
- data retention
- training-data restrictions
- permission enforcement
- action authority
- monitoring
- cost controls

AI output must be clearly distinguished from authoritative human-entered or verified records.

AI must not receive broader access than the initiating user is authorized to exercise.

27. Failure, Recovery, and Correction Standard

For every material failure mode, the PIA must define:

- cause
- detection
- user-visible message
- system state
- data integrity impact
- retry behavior
- rollback behavior
- escalation
- monitoring
- audit evidence
- support procedure
- recovery objective
- correction authority

The PIA must address:

- invalid input
 - duplicate records
 - stale data
 - concurrent edits
 - unauthorized access
 - dependency outage
 - failed integration
 - failed payment
 - failed upload
 - failed notification
 - failed synchronization
 - partial processing
 - background-job failure
 - timeout
 - data corruption
 - deleted or deactivated actor
 - expired authority
 - facility transfer
 - horse transfer
 - time-zone error
 - configuration error
-

28. Nonfunctional Requirements

Every PIA must define measurable nonfunctional requirements for applicable areas:

- availability
- performance
- scalability
- accessibility
- resilience
- recoverability
- maintainability
- portability
- compatibility
- security
- privacy
- observability
- usability
- mobile responsiveness
- offline operation
- data integrity
- backup
- retention
- latency
- throughput
- concurrency
- cost constraints

“Fast,” “scalable,” or “reliable” are not acceptable without measurable thresholds or an approved measurement plan.

29. Acceptance Criteria Standard

Acceptance criteria must be objective and reproducible.

Each criterion must identify:

- criterion ID
- linked requirement
- actor
- precondition
- action
- expected outcome
- prohibited outcome
- audit expectation
- failure expectation
- release scope

- evidence requirement

Preferred structure:

Given [precondition],
When [actor or system action],
Then [required outcome],
And [security, audit, notification, or data condition],
But not [prohibited outcome].

Acceptance criteria must include positive and negative behavior.

30. Test and Validation Standard

Each PIA must include tests covering:

- unit behavior
- service behavior
- API contracts
- database constraints
- state transitions
- permissions
- cross-domain integration
- UI workflows
- mobile workflows
- offline behavior
- synchronization
- accessibility
- performance
- security
- recovery
- migration
- regression
- golden paths
- negative paths
- misuse and abuse cases

Each test must link to one or more requirements and acceptance criteria.

A passed test without preserved evidence does not complete the evidence requirement.

31. Golden-Path Reproduction Standard

Every PIA must define representative end-to-end scenarios.

Each golden path must include:

- realistic actor
- realistic data
- starting state
- complete workflow
- expected state transitions
- expected records
- expected notifications
- expected audit events
- expected cross-domain behavior
- cleanup or completion state
- evidence capture instructions

Golden paths must reflect actual barn operations rather than artificially simple demonstrations.

At least one golden path must cover the first-user enrollment relevance of the domain.

32. Adversarial and Negative Testing Standard

Every PIA must identify how the capability could:

- fail
- be misunderstood
- be misused
- be abused
- expose data
- create conflicting records
- bypass permissions
- generate duplicate actions
- create silent data loss
- mislead a user
- harm a minor
- create financial error
- create health or welfare risk
- create legal or evidentiary ambiguity
- fail under poor connectivity
- fail during rapid repeated actions

The adversarial review must challenge:

- assumptions
- boundary conditions
- default settings
- role escalation
- cross-facility access

- expired relationships
 - transferred horses
 - deleted users
 - stale permissions
 - automation authority
 - administrator access
 - support access
-

33. Evidence Standard

Evidence must be sufficient to show what was tested, against which version, with what result.

Evidence may include:

- automated test output
- screenshots
- screen recordings
- API traces
- database query results
- schema-validation output
- permission-matrix results
- accessibility reports
- security scan results
- migration reports
- synchronization logs
- deployment records
- configuration snapshots
- checksum manifests
- golden-path reproduction records
- adversarial review records
- incident simulations
- backup and restore evidence
- release disposition

Each evidence item must include:

- evidence ID
- date
- producer
- environment
- code version
- PIA version
- linked requirements
- linked tests
- result
- limitations

- checksum or integrity reference where appropriate
- custodian
- retention location

Evidence must not contain unredacted secrets or unnecessary sensitive data.

34. Finding Severity

Findings must use the following severity model:

Severity	Meaning
P0	Immediate safety, security, legal, data-integrity, or release-blocking defect.
P1	Material defect that blocks the applicable approval, implementation, verification, or enrollment gate.
P2	Nonblocking weakness, improvement, or retained risk that must be tracked.
P3	Editorial, cosmetic, or low-impact improvement.

A PIA cannot be:

- implementation-authorized with an unresolved relevant P0;
- enrollment-ready with an unresolved relevant P0 or P1;
- represented as fully conforming when material P2 risks are hidden.

35. Release Classification

Every capability and requirement must be classified as one of:

- REQUIRED_FOR_INTERNAL_BUILD
- REQUIRED_FOR_FOUNDER_PILOT
- REQUIRED_FOR_FIRST_USER_ENROLLMENT
- REQUIRED_BEFORE_PAID_ENROLLMENT
- REQUIRED_FOR_GENERAL_PRODUCTION
- POST_LAUNCH_PLANNED
- DEFERRED_WITH_APPROVAL
- EXPERIMENTAL
- OUT_OF_SCOPE
- PROHIBITED

Deferred work must identify:

- approving authority

- rationale
 - risk
 - dependency
 - target milestone
 - effect on users
 - effect on enrollment readiness
-

36. Dependency Standard

Each dependency must identify:

- dependency ID
- supplying PIA or service
- required capability
- required version
- blocking status
- owner
- interface
- fallback behavior
- verification method
- evidence
- due milestone

Circular dependencies must be resolved before implementation authorization unless an approved staged implementation plan exists.

37. Operational Readiness

Before a PIA can support first-user enrollment, it must define:

- production owner
- support owner
- user support workflow
- administrative tools
- monitoring
- alerts
- incident severity
- escalation
- backup
- restore
- rollback
- release procedure
- feature-flag procedure
- status communication

- known limitations
- onboarding documentation
- user training
- support documentation
- data correction procedure
- account suspension procedure
- access revocation
- offboarding
- retention and deletion operations

A functional feature without support and recovery controls is not operationally complete.

38. Five Mandatory Readiness Questions

Every PIA must answer the following questions with evidence.

Question 1

Can engineering build the capability without making unauthorized product decisions?

Required answer components:

- resolved material requirements
- complete workflows
- explicit business rules
- defined data ownership
- defined permissions
- resolved dependencies
- identified open decisions

Question 2

Can quality assurance determine objectively whether the capability works?

Required answer components:

- measurable acceptance criteria
- test matrix
- negative tests
- failure tests
- golden paths
- evidence requirements

Question 3

Can a reviewer trace the capability to EquineSync's controlling governance and the MAIP?

Required answer components:

- source register
- requirement traceability
- decision register
- forward and backward links
- version baseline

Question 4

Can EquineSync safely operate, support, monitor, recover, and maintain the capability?

Required answer components:

- observability
- support procedures
- recovery
- rollback
- administrative tools
- incident handling
- maintenance ownership

Question 5

Can the Founder determine whether the capability is ready for first-user enrollment?

Required answer components:

- release classification
- enrollment-specific gates
- unresolved findings
- retained risks
- evidence package
- readiness disposition

Permitted answers are:

- YES_WITH_EVIDENCE
- NO
- PARTIALLY_SATISFIED
- NOT_APPLICABLE_WITH_JUSTIFICATION

A PIA is not implementation-ready unless Questions 1 through 3 are answered YES_WITH_EVIDENCE .

A PIA is not ready for first-user enrollment unless all five are answered YES_WITH_EVIDENCE .

39. Formal Deviation Process

A departure from this Standard requires a controlled deviation record.

Each deviation must include:

- deviation ID
- affected requirement
- requested departure
- reason
- alternatives considered
- security impact
- privacy impact
- operational impact
- implementation impact
- testing impact
- user impact
- duration
- compensating control
- risk owner
- approval authority
- expiration or review date

No deviation may override locked constitutional governance without proper constitutional change authority.

40. Required PIA Package

Each PIA package should contain:

```
/[pia-domain]/
  PIA_[DOMAIN]_Vx.y.z.md
  PIA_[DOMAIN]_Vx.y.z.docx
  PIA_[DOMAIN]_MACHINE_READABLE.json
  SOURCE_REGISTER.md
  REQUIREMENT_REGISTER.csv
  DECISION_REGISTER.md
  WORKFLOW_REGISTER.md
  DATA_DICTIONARY.md
  STATE_TRANSITION_MATRIX.csv
  PERMISSION_MATRIX.csv
  API_EVENT_CONTRACTS.md
  ACCEPTANCE_CRITERIA.csv
  TEST_MATRIX.csv
  GOLDEN_PATHS.md
```

```
ADVERSARIAL_SCENARIOS.md
DEPENDENCY_REGISTER.csv
RISK_AND_FINDING_REGISTER.csv
EVIDENCE_MANIFEST.json
CHANGE_CONTROL_LOG.md
REVIEW_DISPOSITION.md
```

Equivalent repository-standard formats may be used when approved.

The human-readable and machine-readable versions must remain materially consistent.

41. Machine-Readable Companion

Each PIA must have a machine-readable companion containing, at minimum:

```
{
  "pia_id": "",
  "title": "",
  "version": "",
  "status": "",
  "classification": "",
  "owner": "",
  "source_baseline": [],
  "requirements": [],
  "workflows": [],
  "entities": [],
  "state_models": [],
  "permission_rules": [],
  "acceptance_criteria": [],
  "tests": [],
  "evidence": [],
  "dependencies": [],
  "risks": [],
  "findings": [],
  "release_scope": [],
  "readiness_questions": [],
  "approvals": [],
  "change_history": []
}
```

Machine validation must confirm:

- required fields are present
- identifiers are unique

- references resolve
 - no test is orphaned
 - no evidence item is orphaned
 - mandatory sections exist
 - release classifications are valid
 - findings use valid severity
 - readiness answers use valid dispositions
-

42. Review Protocol

A complete PIA review must include:

1. Source verification
2. Structural completeness review
3. Domain review
4. Cross-domain boundary review
5. Security and privacy review
6. Permission review
7. Data-integrity review
8. Architecture review
9. Operational-readiness review
10. Adversarial challenge
11. Machine validation
12. Golden-path reproduction review
13. Evidence-plan review
14. Founder decision review

Each review must produce:

- reviewer identity or function
 - scope
 - date
 - version reviewed
 - findings
 - limitations
 - disposition
 - required remediation
 - retained risks
-

43. Implementation Authorization Gate

Founder approval of a PIA does not automatically authorize implementation unless the disposition expressly states implementation is authorized.

Minimum implementation authorization criteria:

- controlling sources verified
- mandatory sections complete
- Questions 1 through 3 answered YES_WITH_EVIDENCE
- no unresolved P0
- no implementation-blocking P1
- dependencies identified
- release scope defined
- acceptance criteria complete
- test strategy complete
- evidence plan complete
- change control active

Recommended disposition:

FOUNDER_APPROVED_AND_IMPLEMENTATION_AUTHORIZED

44. First-User Enrollment Gate

A PIA may support first-user enrollment only when:

- all five readiness questions are YES_WITH_EVIDENCE
- required enrollment capabilities are implemented
- tests passed
- golden paths reproduced
- relevant adversarial scenarios passed or accepted
- evidence is preserved
- operational ownership is assigned
- support and recovery are ready
- no unresolved P0 or P1 remains
- P2 risks are disclosed and accepted
- onboarding materials exist
- monitoring is active
- rollback capability is verified
- Founder issues an enrollment-readiness disposition

Recommended disposition:

VERIFIED_AND_READY_FOR_CONTROLLED_FIRST_USER_ENROLLMENT

45. Prohibited PIA Anti-Patterns

A PIA is deficient when it:

- merely lists features
- copies governance without translating it into implementation
- describes only screens
- omits negative paths
- treats admins as unlimited
- fails to define data ownership
- leaves permissions to engineering
- uses vague acceptance criteria
- says “offline supported” without behavior
- says “secure” without controls
- says “AI-powered” without authority and review limits
- omits support and recovery
- omits evidence requirements
- hides unresolved decisions
- labels deferred features as complete
- mixes authoritative and derived data
- creates competing sources of truth
- relies on UI hiding instead of authorization
- treats a passing demo as verification
- declares readiness without preserved evidence
- silently changes a locked governance requirement

PART II

CONTROLLED PRODUCT IMPLEMENTATION ATLAS TEMPLATE

[PIA TITLE]

PIA ID: [ES-PIA-DOMAIN-Vx.y.z]

Version: [x.y.z]

Status: [LIFECYCLE_STATUS]

PIA Type: [FOUNDATIONAL / DOMAIN / CROSS-DOMAIN / PLATFORM / INTEGRATION /
EXPERIENCE / MIGRATION / ASSURANCE]

Classification: EquineSync Internal

Founder / Approval Authority: Rian Ray

PIA Owner: [NAME OR FUNCTION]
Engineering Owner: [NAME OR FUNCTION]
Operational Owner: [NAME OR FUNCTION]
Evidence Custodian: [NAME OR FUNCTION]
Effective Date: [DATE OR PENDING]
Constitutional Baseline: [COMMIT / TAG]
MAIP Baseline: [VERSION / PATH / HASH]
Predecessor: [DOCUMENT OR NONE]
External Assurance: NOT_EXTERNALLY_ASSURED
Release Applicability: [INTERNAL / PILOT / FIRST USER / PAID / GENERAL PRODUCTION]

1. Document Control and Status

1.1 Document Control

Field	Value
PIA ID	
Title	
Version	
Status	
PIA Type	
Owner	
Drafting Function	
Reviewers	
Approval Authority	
Source Baseline	
Repository Path	
Effective Date	
Supersedes	
External Assurance	

1.2 Current Disposition

[CONTROLLED_DISPOSITION]

1.3 Implementation Status

Area	Status	Evidence
Product requirements		
Workflows		
Data model		
Permissions		
APIs and events		
User interfaces		
Testing		
Evidence		
Operations		
Enrollment readiness		

2. Executive Summary

Provide a concise explanation of:

- the domain
- the primary users
- the capability being implemented
- the operational problem being solved
- the release scope
- the highest-risk areas
- the current readiness status

2.1 Executive Readiness Summary

Question	Answer	Evidence
Buildable without unauthorized decisions?		
Objectively testable?		
Traceable to governance and MAIP?		
Safely operable and recoverable?		
Ready for first-user enrollment?		

3. Purpose and Intended Outcome

3.1 Purpose

[State why this PIA exists.]

3.2 Intended Product Outcome

[Describe the user and business outcome.]

3.3 Intended Operational Outcome

[Describe the operational result.]

3.4 Success Definition

[Define measurable success.]

3.5 Non-Goals

[List what this PIA does not attempt to accomplish.]

4. Authoritative Sources

4.1 Source Register

Source ID	Source	Type	Version / Hash	Authority	Requirements Derived
-----------	--------	------	----------------	-----------	----------------------

4.2 Founder Decisions

Decision ID	Decision	Date	Status	Affected Requirements
-------------	----------	------	--------	-----------------------

4.3 Assumptions

Assumption ID	Assumption	Basis	Risk	Validation Plan
---------------	------------	-------	------	-----------------

4.4 Source Gaps

Gap ID	Missing Source or Decision	Effect	Blocking?	Owner
--------	----------------------------	--------	-----------	-------

5. Domain Scope and Boundaries

5.1 In Scope

[List included capabilities.]

5.2 Out of Scope

[List excluded capabilities.]

5.3 Domain Ownership

Record / Decision	Ownership Type	Authoritative Domain	Steward	Notes
-------------------	----------------	----------------------	---------	-------

5.4 Cross-Domain Boundary Matrix

Related Domain	This PIA Owns	This PIA References	Prohibited Overlap	Integration Method
----------------	---------------	---------------------	--------------------	--------------------

5.5 Boundary Conflicts

[Identify unresolved or historically conflicting boundaries.]

6. Definitions and Controlled Vocabulary

Term	Definition	Source	Prohibited Alternative Meaning
------	------------	--------	--------------------------------

Terms must align with the Constitutional Vocabulary and Definition Index.

7. Actors, Roles, Relationships, and Authorities

7.1 Actor Register

Actor ID	Actor	Description	Authentication Required	Typical Authority
----------	-------	-------------	-------------------------	-------------------

7.2 Relationship Register

Relationship	Parties	Start Condition	End Condition	Authority Created
--------------	---------	-----------------	---------------	-------------------

7.3 Delegation Rules

Delegator	Delegate	Delegable Authority	Duration	Revocation	Audit
-----------	----------	---------------------	----------	------------	-------

7.4 Guardian and Minor Rules

[Define all applicable guardian, minor, consent, communication, and safeguarding requirements.]

8. Capability Map

Capability ID	Capability	Description	Primary Actor	Release Classification	Status
---------------	------------	-------------	---------------	------------------------	--------

8.1 Capability Dependencies

[Provide a capability dependency diagram or ordered list.]

9. User and Operational Workflows

Repeat the following structure for each workflow.

9.X [WORKFLOW ID]: [WORKFLOW NAME]

Purpose:

Primary Actor:

Supporting Actors:

Trigger:

Preconditions:

Required Inputs:

Optional Inputs:

Authorization:

Starting State:

Completion State:

Release Classification:

Workflow Steps

Step	Actor / System	Action	Validation	State Change	Audit Event	Notification
------	----------------	--------	------------	--------------	-------------	--------------

Success Outcome

[Define completion.]

Partial Success

[Define partial processing and user visibility.]

Failure Behavior

[Define error, rollback, retry, and escalation.]

Cancellation and Reversal

[Define whether and how the workflow can be canceled or reversed.]

Offline Behavior

[Define offline availability and synchronization.]

Acceptance Criteria

[List linked acceptance criteria.]

10. Business Rules

Rule ID	Rule	Source	Enforcement Layer	Exception Authority	Test
---------	------	--------	-------------------	---------------------	------

Business rules must be enforced in the authoritative service layer where material.

11. Data Entities and Relationships

11.1 Entity Register

Entity ID	Entity	Purpose	Authoritative Owner	Sensitivity	Retention
-----------	--------	---------	---------------------	-------------	-----------

11.2 Field Definitions

Entity	Field	Type	Required	Validation	Default	Sensitive	Searchable	Offline
--------	-------	------	----------	------------	---------	-----------	------------	---------

11.3 Relationships

Parent Entity	Child Entity	Relationship	Cardinality	Deletion Behavior
---------------	--------------	--------------	-------------	-------------------

11.4 Provenance and Derived Data

Field / Record	Classification	Source	Derivation	Human Review	Correction
----------------	----------------	--------	------------	--------------	------------

12. Record Ownership and Stewardship

Record Type	Authoritative Owner	Steward	Creator	Editor	Corrector	Deletion Authority	Audit
-------------	---------------------	---------	---------	--------	-----------	--------------------	-------

12.1 Correction Rules

[Define correction without destruction of historical truth.]

12.2 Transfer Rules

[Define ownership, facility, horse, user, or relationship transfer effects.]

13. State and Transition Models

13.1 State Register

State ID	State	Meaning	Entry Requirements	Exit Conditions	Terminal?
----------	-------	---------	--------------------	-----------------	-----------

13.2 Transition Matrix

From State	Trigger	Authorized Actor	Required Data	To State	Audit	Notification	Reversible
------------	---------	------------------	---------------	----------	-------	--------------	------------

13.3 Prohibited Transitions

[List prohibited transitions and enforcement.]

14. Authorization and Permission Matrix

Permission ID	Actor / Role	Relationship	Resource	Action	Conditions	Delegate?	Audit	Notification
---------------	--------------	--------------	----------	--------	------------	-----------	-------	--------------

14.1 Platform Administrator Boundary

[Define what platform administrators may and may not access or modify.]

14.2 Emergency Access

[Define whether emergency access exists and its controls.]

14.3 Revocation

[Define immediate and delayed revocation behavior.]

15. User Interface and Experience Requirements

15.1 Supported Surfaces

Surface	Support Level	Primary Uses	Limitations
Web			
iOS			
Android			
Owner Portal			
Admin Portal			
Public Enrollment			

15.2 Screen and Interaction Register

UI ID	Screen / Component	Actor	Purpose	Required States	Accessibility
-------	--------------------	-------	---------	-----------------	---------------

15.3 Required Interaction States

For each material screen define:

- initial
- loading
- empty
- populated
- validation error
- permission denied
- offline
- sync pending
- sync failed

- success
- partial success
- destructive confirmation

15.4 Accessibility

[Define keyboard, screen reader, contrast, touch target, motion, language, and cognitive accessibility requirements.]

16. API, Event, Job, and Integration Contracts

16.1 API Register

API ID	Method / Operation	Purpose	Caller	Authorization	Idempotent	Version
--------	--------------------	---------	--------	---------------	------------	---------

16.2 Event Register

Event ID	Event	Producer	Consumers	Payload	Delivery Guarantee	Audit
----------	-------	----------	-----------	---------	--------------------	-------

16.3 Job Register

Job ID	Job	Trigger	Schedule	Retry	Failure Escalation	Idempotency
--------	-----	---------	----------	-------	--------------------	-------------

16.4 Integration Register

Integration ID	External System	Purpose	Data Exchanged	Authority	Failure Behavior
----------------	-----------------	---------	----------------	-----------	------------------

17. Notifications and Communications

Notification ID	Trigger	Recipient	Channel	Required Content	Timing	Retry	Audit
-----------------	---------	-----------	---------	------------------	--------	-------	-------

17.1 Preferences

[Define user-configurable and mandatory notifications.]

17.2 Delivery Failure

[Define user and support visibility.]

18. Files, Media, and Document Handling

Asset Type	Allowed Formats	Size Limit	Owner	Access	Retention	Malware Scan	Export
------------	-----------------	------------	-------	--------	-----------	--------------	--------

Define:

- upload
- preview
- replacement
- versioning
- deletion
- redaction
- sharing
- download
- metadata
- evidence preservation

19. Search, Reporting, and Analytics Requirements

19.1 Search

Search Requirement	Indexed Data	Permission Filter	Ranking	Offline	Audit
--------------------	--------------	-------------------	---------	---------	-------

19.2 Reports

Report	Audience	Source Data	Freshness	Export	Permission
--------	----------	-------------	-----------	--------	------------

19.3 Analytics

[Define permitted metrics, aggregation, privacy, and prohibited inferences.]

20. Offline and Synchronization Behavior

Workflow / Record	Offline Read	Offline Write	Queue	Conflict Rule	User Status	Max Offline Duration
-------------------	--------------	---------------	-------	---------------	-------------	----------------------

20.1 Conflict Resolution

[Define server wins, client wins, merge, human review, or domain-specific resolution.]

20.2 Device Security

[Define local encryption, revocation, cache expiration, and logout behavior.]

21. Security, Privacy, Consent, and Safeguarding

21.1 Data Classification

Data	Classification	Access Basis	Encryption	Retention	Export Restriction
------	----------------	--------------	------------	-----------	--------------------

21.2 Security Controls

[List authentication, authorization, session, rate-limit, abuse, and administrative controls.]

21.3 Privacy Controls

[List minimization, purpose limitation, consent, access, correction, retention, and deletion controls.]

21.4 Safeguarding Controls

[List minor, guardian, communication, location, media, reporting, and escalation controls.]

21.5 Threat and Abuse Cases

Threat	Asset	Actor	Control	Detection	Response
--------	-------	-------	---------	-----------	----------

22. Failure Modes, Recovery, and Correction

Failure ID	Failure	Detection	User Experience	System State	Retry	Recovery	Escalation
------------	---------	-----------	-----------------	--------------	-------	----------	------------

22.1 Data Correction

[Define correction process and authority.]

22.2 Rollback

[Define rollback scope and evidence.]

22.3 Disaster Recovery

[Define backup, restore, recovery objectives, and test requirements.]

23. Observability, Administration, and Support Operations

23.1 Metrics

Metric ID	Metric	Purpose	Threshold	Alert	Owner
-----------	--------	---------	-----------	-------	-------

23.2 Logs and Audit

Event	Log Type	Retention	Sensitive Data Handling	Support Use
-------	----------	-----------	-------------------------	-------------

23.3 Administrative Tools

[List tools required for support, correction, suspension, replay, reconciliation, and diagnostics.]

23.4 Support Procedures

[Define intake, triage, escalation, user communication, and closure.]

24. Nonfunctional Requirements

NFR ID	Category	Requirement	Threshold	Validation	Release Scope
--------	----------	-------------	-----------	------------	---------------

Include:

- performance
- availability
- accessibility
- scalability
- resilience
- backup
- recovery
- maintainability
- compatibility
- security
- usability
- offline operation

25. Migration, Configuration, Feature Flags, and Seed Data

25.1 Migration

Migration ID	Source	Target	Mapping	Validation	Rollback	Evidence
--------------	--------	--------	---------	------------	----------	----------

25.2 Configuration

Setting	Scope	Default	Allowed Values	Change Authority	Audit
---------	-------	---------	----------------	------------------	-------

25.3 Feature Flags

Flag	Purpose	Default	Environment	Owner	Removal Condition
------	---------	---------	-------------	-------	-------------------

25.4 Seed Data

[List required roles, permissions, categories, templates, statuses, and reference data.]

26. Acceptance Criteria

AC ID	Requirement	Given	When	Then	But Not	Evidence
-------	-------------	-------	------	------	---------	----------

27. Test and Validation Matrix

Test ID	Requirement	Test Type	Environment	Preconditions	Expected Result	Evidence ID	Status
---------	-------------	-----------	-------------	---------------	-----------------	-------------	--------

Include:

- positive
- negative
- permission
- state
- integration
- mobile
- offline
- performance
- accessibility
- security

- recovery
- migration
- regression

28. Golden-Path Reproduction Scenarios

28.1 [GOLDEN PATH ID]: [NAME]

Purpose:

Actors:

Starting State:

Data Set:

Steps:

Expected States:

Expected Records:

Expected Notifications:

Expected Audit Events:

Expected Completion:

Evidence Required:

First-User Enrollment Relevance:

Repeat for each golden path.

29. Adversarial, Negative, and Abuse Scenarios

Scenario ID	Scenario	Risk	Expected Defense	Detection	Evidence
-------------	----------	------	------------------	-----------	----------

Include:

- unauthorized access
- expired authority
- cross-facility access
- duplicate action
- race condition
- stale data
- offline conflict
- malicious upload
- notification abuse
- support misuse
- admin misuse
- AI misuse
- financial manipulation
- record tampering

30. Evidence Requirements and Evidence Manifest

Evidence ID	Requirement / Test	Evidence Type	Producer	Environment	Integrity Method	Custodian	Status
-------------	--------------------	---------------	----------	-------------	------------------	-----------	--------

30.1 Evidence Package Requirements

[List naming, checksums, storage, redaction, retention, and packaging rules.]

31. Dependencies and Implementation Sequence

31.1 Dependency Register

Dependency ID	Dependency	Supplier	Blocking	Required Version	Verification	Status
---------------	------------	----------	----------	------------------	--------------	--------

31.2 Implementation Sequence

Order	Work Package	Preconditions	Output	Review Gate
-------	--------------	---------------	--------	-------------

31.3 Critical Path

[Identify the critical implementation path.]

32. Release Scope and Readiness Gates

32.1 Release Classification

Capability / Requirement	Classification	Included?	Deferral Authority	Risk
--------------------------	----------------	-----------	--------------------	------

32.2 Internal Build Gate

[List requirements.]

32.3 Founder Pilot Gate

[List requirements.]

32.4 First-User Enrollment Gate

[List requirements.]

32.5 Paid Enrollment Gate

[List requirements.]

32.6 General Production Gate

[List requirements.]

33. Open Decisions, Findings, Assumptions, and Retained Risks

33.1 Open Decisions

OQ ID	Question	Impact	Owner	Due Gate	Status
-------	----------	--------	-------	----------	--------

33.2 Findings

Finding ID	Severity	Finding	Affected Requirement	Required Action	Status
------------	----------	---------	----------------------	-----------------	--------

33.3 Risks

Risk ID	Risk	Likelihood	Impact	Control	Owner	Acceptance
---------	------	------------	--------	---------	-------	------------

33.4 Deviations

Deviation ID	Requirement	Departure	Compensating Control	Approval	Expiration
--------------	-------------	-----------	----------------------	----------	------------

34. Change-Control History

Version	Date	Change	Reason	Requirements Affected	Approved By
---------	------	--------	--------	-----------------------	-------------

35. Requirement Traceability Matrix

Source	Requirement	Workflow	Entity	Permission	AC	Test	Evidence	Release Gate
--------	-------------	----------	--------	------------	----	------	----------	--------------

Every row must resolve in both directions.

36. Five Mandatory Readiness Questions

36.1 Engineering Buildability

Question: Can engineering build the capability without making unauthorized product decisions?

Answer: [YES_WITH_EVIDENCE / NO / PARTIALLY_SATISFIED]

Evidence and Explanation:

[Provide requirement, workflow, data, permission, and dependency evidence.]

36.2 Objective QA Verification

Question: Can quality assurance determine objectively whether the capability works?

Answer: [YES_WITH_EVIDENCE / NO / PARTIALLY_SATISFIED]

Evidence and Explanation:

[Provide acceptance criteria, tests, negative scenarios, and evidence plan.]

36.3 Governance and MAIP Traceability

Question: Can a reviewer trace the capability to EquineSync's controlling governance and the MAIP?

Answer: [YES_WITH_EVIDENCE / NO / PARTIALLY_SATISFIED]

Evidence and Explanation:

[Provide source and traceability evidence.]

36.4 Operational Safety and Recovery

Question: Can EquineSync safely operate, support, monitor, recover, and maintain the capability?

Answer: [YES_WITH_EVIDENCE / NO / PARTIALLY_SATISFIED]

Evidence and Explanation:

[Provide operational, support, observability, recovery, and rollback evidence.]

36.5 First-User Enrollment Readiness

Question: Can the Founder determine whether the capability is ready for first-user enrollment?

Answer: [YES_WITH_EVIDENCE / NO / PARTIALLY_SATISFIED]

Evidence and Explanation:

[Provide release scope, findings, risks, test results, evidence, and readiness disposition.]

37. Review, Approval, and Implementation Authorization

37.1 Review Record

Review Function	Reviewer	Version	Date	Disposition	Findings
Domain review					
Architecture review					
Security and privacy review					
Segregated review					
Adversarial challenge					
Machine validation					
Golden-path review					
Evidence review					

37.2 Founder Decision

Founder: Rian Ray

Decision Date:

Disposition:

Select one:

- RETURNED_FOR_REVISION
- FOUNDER_APPROVED_NOT_IMPLEMENTATION_AUTHORIZED
- FOUNDER_APPROVED_AND_IMPLEMENTATION_AUTHORIZED
- FOUNDER_APPROVED_WITH_RETAINED_NONBLOCKING_P2
- IMPLEMENTATION_VERIFIED_WITH_OPEN_FINDINGS
- VERIFIED_RELEASE_CANDIDATE
- VERIFIED_AND_READY_FOR_CONTROLLED_FIRST_USER_ENROLLMENT
- NOT_READY_FOR_FIRST_USER_ENROLLMENT
- SUPERSEDED

- **WITHDRAWN**

37.3 Conditions

[List conditions, retained risks, required follow-up, or limitations.]

37.4 Signature Record

Authority	Name	Decision	Date
Founder	Rian Ray		
PIA Owner			
Engineering Owner			
Evidence Custodian			

PART III

PIA READINESS CHECKLIST

A PIA is ready for Founder review only when all applicable items below are satisfied.

A. Authority and Sources

- ☐ Controlling sources are identified.
- ☐ Source versions and hashes are recorded where available.
- ☐ Founder decisions are included.
- ☐ No material source conflict is concealed.
- ☐ Assumptions are identified.
- ☐ Missing sources are classified.

B. Scope and Boundaries

- ☐ In-scope capabilities are explicit.
- ☐ Out-of-scope capabilities are explicit.
- ☐ Domain ownership is clear.
- ☐ Cross-domain boundaries are defined.
- ☐ Competing sources of truth are prohibited.
- ☐ Dependencies are identified.

C. Product and Workflow

- ☐ Actors are defined.

- ☐ Relationships are defined.
- ☐ Workflows include start through completion.
- ☐ Negative paths are defined.
- ☐ Cancellation and reversal are defined.
- ☐ Notifications are defined.
- ☐ Offline behavior is defined.

D. Data and Permissions

- ☐ Entities and fields are defined.
- ☐ Record ownership is defined.
- ☐ Provenance is defined.
- ☐ Derived and AI-generated data are distinguished.
- ☐ Permission rules are explicit.
- ☐ Delegation and revocation are defined.
- ☐ Guardian and minor rules are addressed.

E. Technical Implementation

- ☐ APIs are defined.
- ☐ Events are defined.
- ☐ Jobs are defined.
- ☐ Integrations are defined.
- ☐ Idempotency is addressed.
- ☐ Retry and timeout behavior are addressed.
- ☐ Configuration and feature flags are defined.

F. Security and Operations

- ☐ Authentication requirements are defined.
- ☐ Authorization is enforced beyond the UI.
- ☐ Privacy and consent are addressed.
- ☐ Safeguarding is addressed.
- ☐ Failure modes are defined.
- ☐ Monitoring and alerts are defined.
- ☐ Support and administrative tools are defined.
- ☐ Backup, restore, and rollback are defined.

G. Verification and Evidence

- ☐ Acceptance criteria are objective.
- ☐ Tests map to requirements.
- ☐ Positive and negative tests exist.
- ☐ Golden paths exist.
- ☐ Adversarial scenarios exist.
- ☐ Evidence requirements are defined.
- ☐ Evidence custody is assigned.

- [] No orphaned requirement, test, or evidence item exists.

H. Release Readiness

- [] Release classifications are assigned.
 - [] First-user enrollment requirements are identified.
 - [] Deferred work is approved.
 - [] No unresolved P0 remains.
 - [] No blocking P1 remains.
 - [] Retained P2 risks are disclosed.
 - [] The five mandatory readiness questions are answered.
 - [] The requested Founder disposition is stated.
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PART IV

APPROVAL EFFECT

Upon Founder adoption, this Standard becomes the controlling drafting and review standard for all EquineSync Product Implementation Atlases.

No EquineSync PIA may be represented as:

- complete,
- implementation-ready,
- implementation-authorized,
- verified,
- release-ready, or
- ready for first-user enrollment

unless it satisfies the requirements and applicable gates established by this Standard or has a valid Founder-authorized deviation.

This Standard is intended to preserve the integrity of the full implementation chain:

Governance establishes authority.
The MAIP establishes coordinated implementation direction.
The PIA establishes domain-level executable truth.
Engineering creates the implementation.
Testing challenges the implementation.
Evidence proves the result.
The Founder determines whether enrollment may begin.